The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Primitive Substrate Shell Restriction
Next Benchmark Criterion

Abstract

The current split-core routing note had fixed the honest burden below the selector-law collapse frontier: first derive or uniquely force the split zero-hidden selector-shell core $\mathfrak{C}^{\mathrm{sel,split}}$ from still more primitive explicit substrate structure, or else prove a genuinely smaller theorem inside that split core. The new shell-restriction forcing theorem has now spent that first admissible route. The split core $\mathfrak{C}^{\mathrm{sel,split}}$ is no longer left as an independently inserted stopping point; it is a canonical and unique output of the primitive substrate shell-restriction layer $\mathfrak{S}^{\mathrm{sub,sh}}$.

This note records the routing consequence of that new theorem. The next honest benchmark-facing manuscript must therefore go one level below shell-restriction scope. Its first try should be to derive or uniquely force the shell-generating substrate lift package $\mathfrak{L}^{\mathrm{sub,lift}}$ from still more primitive explicit substrate structure in a way that actually changes benchmark-facing status. If that cannot be done cleanly, the legitimate fallback is a genuinely smaller theorem-level collapse, sharper necessity result, or sharper uniqueness result strictly inside the shell-restriction layer $\mathfrak{S}^{\mathrm{sub,sh}}$ itself. Another split-core relay, selector-law restatement, combined-response restatement, ambient-package restatement, trace-core restatement, completion-core restatement, exact-shell affine nucleus restatement, or benchmark-neutral shell-restriction relay that preserves the same shell-restriction package without shrinking or forcing it further does not count next.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and stronger first-principles promotion language remains paused. The positive result of this note is therefore routing discipline at the new shell-restriction frontier.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity is no longer at a stage where the split zero-hidden selector-shell core $\mathfrak{C}^{\mathrm{sel,split}}$ should be treated as the default unexplained stopping point. The split-core collapse theorem had already shown that the full selector-law presentation carries no extra benchmark-facing freedom beyond $\mathfrak{C}^{\mathrm{sel,split}}$. The new shell-restriction forcing theorem then spent the first admissible upstream option below that frontier: $\mathfrak{C}^{\mathrm{sel,split}}$ is itself a canonical and unique output of the primitive substrate shell-restriction layer $\mathfrak{S}^{\mathrm{sub,sh}}$.

The present note records the routing consequence of that theorem. It does not reopen the split-core mathematics, and it does not claim a completed first-principles substrate derivation of gravity. It records only which kind of next manuscript would now count as an honest benchmark-facing gain below the shell-restriction frontier.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It records only which kind of next manuscript would now count as an honest benchmark-facing gain below the shell-restriction frontier.

2. Fixed Inputs

Definition 1 (Shell-restriction-frontier fixed inputs)**.** *The criterion recorded here uses the following fixed inputs.*

1. *The benchmark exact-closure package, which fixes the public one-metric GR target of the repository.*
2. *The benchmark gatekeeping layer, including the post-bridge status correction and the upstream-compression safeguard that blocks benchmark-neutral deeper-origin relays from counting as new front-line progress once the benchmark-relevant object is unchanged.*
3. *The split-core collapse theorem, which proves that the full zero-hidden selector law carries no extra benchmark-facing freedom beyond the smaller split selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$*.*
4. *The new shell-restriction forcing theorem, which proves that the split selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$ *is itself a canonical and unique output of the primitive substrate shell-restriction layer* $\mathfrak{S}^{\mathrm{sub,sh}}$*.*
5. *The preserved direct-support shell-restriction theorem, which records that* $\mathfrak{S}^{\mathrm{sub,sh}}$ *is itself the structured output of a deeper shell-generating substrate lift package* $\mathfrak{L}^{\mathrm{sub,lift}}$ *on the same one-metric line.*

Remark 1. *The present note does not replace those manuscripts. It records the routing consequence of reading them together under the active safeguard against recursive depth drift.*

3. Why Shell Restriction Is Now the Frontier

Definition 2 (Benchmark-neutral shell-restriction relay)**.** *A* benchmark-neutral shell-restriction relay *is a manuscript that preserves the same benchmark one-metric output, the same ordinary matter route, the same split selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$*, the same selector-law consequence, the same combined-response consequence, the same ambient-package consequence, the same trace-core consequence, the same completion-core consequence, and the same claim boundary while merely restating* $\mathfrak{C}^{\mathrm{sel,split}}$*, the selector law, the combined response law, the ambient package, the trace core, the completion core, the exact-shell affine nucleus, or the same shell-restriction package* $\mathfrak{S}^{\mathrm{sub,sh}}$ *without shrinking or forcing* $\mathfrak{S}^{\mathrm{sub,sh}}$ *further.*

Proposition 1 (The shell-restriction forcing theorem is the live frontier)**.** *At the current stage of the primitive-reservoir observer-reduction line, the theorem forcing the split selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$ *from the primitive substrate shell-restriction layer* $\mathfrak{S}^{\mathrm{sub,sh}}$ *is the live benchmark-facing frontier.*

Proof. That theorem changes benchmark-facing status at current scope because it removes $\mathfrak{C}^{\mathrm{sel,split}}$ itself as the live internal stopping point. The split core is no longer merely the smaller datum singled out inside the selector law; it is now forced by the deeper shell-restriction layer. By contrast, a manuscript that preserves the same shell-restriction package and the same split-core consequence while merely restating $\mathfrak{C}^{\mathrm{sel,split}}$ or another already-forced larger or smaller bridge object does not alter benchmark-facing status unless it adds a comparably sharp gain. Therefore the live frontier is the shell-restriction forcing theorem rather than the earlier split-core collapse theorem or a later same shell-restriction relay. □

Corollary 1 (Status of the split core as a next-step target)**.** *Under the current routing safeguard, the split zero-hidden selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$ *is no longer itself the default next benchmark-facing target.*

Proof. The shell-restriction forcing theorem already proves that $\mathfrak{C}^{\mathrm{sel,split}}$ is a canonical and unique output of $\mathfrak{S}^{\mathrm{sub,sh}}$. Once that is on record, another manuscript that spends the move on $\mathfrak{C}^{\mathrm{sel,split}}$ again does not remove any further benchmark-relevant freedom. The open burden has therefore moved below split-core scope. □

4. The Next Benchmark Criterion

Definition 3 (Admissible next benchmark-facing shell-restriction manuscript)**.** *At the current shell-restriction frontier, a manuscript counts as an admissible next benchmark-facing move only if it respects the following burden order:*

1. *its primary attempt is to derive or uniquely force the shell-generating substrate lift package* $\mathfrak{L}^{\mathrm{sub,lift}}$ *from still more primitive explicit substrate structure in a way that does more than merely relocate the unexplained layer deeper;*

2. *if that cannot be done cleanly at current scope, its admissible fallback gain is to prove a genuinely smaller theorem-level collapse, sharper necessity result, or sharper uniqueness result strictly inside the shell-restriction layer* $\mathfrak{S}^{\mathrm{sub,sh}}$*;*

3. *in either case it preserves the same benchmark one-metric package, the same ordinary matter route, and the same claim boundary, and it introduces neither a second operative metric nor an enlarged low-energy matter law and licenses no stronger first-principles promotion language.*

Theorem 1 (Next honest benchmark-facing task at shell-restriction scope)**.** *The next honest benchmark-facing manuscript at the current shell-restriction frontier should therefore go one level below that frontier: first try to derive or uniquely force the shell-generating substrate lift package* $\mathfrak{L}^{\mathrm{sub,lift}}$ *from still more primitive explicit substrate structure in a way that actually changes benchmark-facing status. If that cannot be done cleanly, the admissible fallback is to prove a genuinely smaller collapse, sharper necessity result, or sharper uniqueness result strictly inside* $\mathfrak{S}^{\mathrm{sub,sh}}$*. A manuscript that only restates the split selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$*, only restates the selector law, only restates the combined response law, only restates the ambient package, only restates the trace core, only restates*

the full completion-core presentation, only restates the exact-shell affine nucleus, or only repeats the same shell-restriction consequence while preserving $\mathfrak{S}^{\mathrm{sub,sh}}$ *without shrinking or forcing it further does not satisfy the criterion.*

Proof. By Proposition 1, the live frontier already lies at the theorem that forces the split core from shell restriction. Therefore a second manuscript below it must improve on that status rather than merely accompany it. A deeper-origin manuscript can do so only if it changes benchmark-facing status by more than depth alone, and the first clean way to do that is to force the shell-generating substrate lift package that already sits immediately upstream in the direct-support record. If that first option is not yet available cleanly, the routing safeguard allows the fallback of a smaller internal collapse, necessity result, or uniqueness result inside $\mathfrak{S}^{\mathrm{sub,sh}}$. If a manuscript simply reproduces the same shell-restriction consequence through another relay while preserving the same shell-restriction package without shrinking or forcing it further, the routing rule classifies it as benchmark neutral. □

5. What the Next Move Should Not Be

Proposition 2 (Screened next moves)**.** *At the current shell-restriction frontier, none of the following is the default next benchmark-facing move:*

1. *another restatement of the split zero-hidden selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$*;*
2. *another restatement of the full zero-hidden selector law;*
3. *another restatement of the combined response substrate law;*
4. *another restatement of the completion-balance ambient package;*
5. *another restatement of the completion-state shell-trace core;*
6. *another restatement of the full completion-core presentation;*
7. *another restatement of the exact-shell affine nucleus;*
8. *another shell-restriction relay that preserves the same* $\mathfrak{S}^{\mathrm{sub,sh}}$ *without shrinking or forcing it further;*
9. *a manuscript that introduces a second operative metric, enlarges the ordinary matter law, or uses stronger first-principles promotion language.*

Proof. Items (1)–(8) fail the preceding criterion because they do not directly address the current shell-restriction burden. They revisit either already-forced larger bridge objects, already-forced smaller bridge objects, or the same shell-restriction stopping point rather than removing the remaining shell-restriction freedom or proving a still smaller internal datum inside $\mathfrak{S}^{\mathrm{sub,sh}}$. Item (9) violates the fixed claim boundary of the benchmark package and therefore cannot count as a disciplined next step on the active line. The benchmark package remains the exact one-metric GR package already fixed by adoption, so any admissible next manuscript must preserve that boundary while keeping stronger first-principles promotion language paused. □

Remark 2. *This screening statement is not a denial that the selector-law collapse theorem, the split-core forcing theorem, or the preserved shell-restriction support theorem matter historically. It*

is a routing statement: they are not the honest way to spend the next benchmark-facing move from the current shell-restriction frontier.

6. Conclusion

The positive result of this note is routing discipline at the current shell-restriction frontier. The shell-restriction forcing theorem remains the live benchmark-facing gain because it already removes the split selector-shell core as the internal stopping point on the active line. The split core therefore no longer sets the honest next burden, and neither do the already-forced selector law, combined response law, ambient package, trace core, completion-core presentation, or exact-shell affine nucleus.

The scope remains narrow and honest. The benchmark package is unchanged: the one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. Nothing here upgrades adoption to derivation or licenses stronger first-principles promotion language yet.

The next open burden is therefore clear. The next benchmark-facing manuscript should go one level below the current shell-restriction frontier: first try to derive or uniquely force the shell-generating substrate lift package $\mathfrak{L}^{\mathrm{sub,lift}}$ from still more primitive explicit substrate structure in a way that actually changes benchmark-facing status while preserving the same benchmark one-metric package and claim boundary. If that cannot be done cleanly, the admissible fallback is a genuinely smaller collapse, sharper necessity result, or sharper uniqueness result strictly inside the shell-restriction layer $\mathfrak{S}^{\mathrm{sub,sh}}$. Another split-core relay, selector-law restatement, combined-response restatement, ambient-package restatement, trace-core restatement, completion-core restatement, exact-shell affine nucleus restatement, or another shell-restriction relay that preserves the same shell-restriction package without shrinking or forcing it further is not the clean next manuscript target at current scope.